# Point-by-Point Response to Reviewers

Manuscript: CorePAM: a 24-gene PAM50-derived expression score with cross-platform external validation for breast cancer prognosis

Journal: Breast Cancer Research

Round: R2 Revision

## Reviewer 1

We thank the reviewer for their constructive feedback, which has improved the presentation of the manuscript.

### Comment R1.1 — Kaplan-Meier figures

*Figures 2-6 present Kaplan Meier curves for the different datasets. The annotations on the plots report two different p-values. Each plot should report a single p-value clearly linked to the comparison being presented (or, if multiple tests are intentionally reported, the figure legend should explicitly define what each p-value corresponds to). In addition, these figures could be consolidated into a single multi-panel figure (one dataset per panel). For consistency, all panels should use the same color scheme, legend format, and annotation style.*

**Response:**

We thank the reviewer for this constructive suggestion. Figures 2–6 have been consolidated into a single multi-panel Figure 2 (panels A–E, one per cohort). Each panel now reports a single log-rank p-value and the dichotomised Cox HR with 95% CI. All panels share a consistent colour scheme, legend format, and annotation style.

### Comment R1.2 — Supplementary numbering

*Supplementary figures and tables should be numbered in the order in which they are first cited in the text. For example, Figure S2 is currently cited before Figure S1, which is confusing.*

**Response:**

All supplementary materials have been renumbered as Additional files 1–18 following the order of first citation in the text. Cross-references throughout the manuscript were updated accordingly.

### Comment R1.3 — Unreferenced supplementary figures

*All supplementary figures should be referenced in the main text. At present, Figures S3 and S4 are not mentioned.*

**Response:**

All supplementary figures and tables are now explicitly referenced in the main text at the point of first relevance.

### Comment R1.4 — Discussion narrative flow and limitations

*The Discussion lacks narrative flow and largely mirrors the Results section, which makes it feel repetitive. The limitations section, in particular, reads as a list; it would benefit from explaining how the limitations may have affected the analyses/interpretation and, where possible, how they could be addressed in future work.*

**Response:**

We appreciate this feedback and have revised the Discussion to improve narrative flow. The opening paragraph was rewritten to frame the principal finding and its scope rather than restating numerical results. A new paragraph presents the direct external head-to-head comparison between CorePAM and the fuller PAM50-based comparator. The Limitations section was restructured into three interpretive subsections—study design and data, analytical choices, and clinical translation—with commentary on how each limitation may affect interpretation.

## Reviewer 2

### Comment R2 — Tone down claims on cross-platform applicability, clinical utility, and non-inferiority

*This revised manuscript is much improved, but a few issues still need revision: the claims on cross-platform applicability, clinical utility, and "non-inferiority" should be toned down, because external calibration remains suboptimal and the frozen-reference analysis suggests that single-sample performance is not yet stable across all platforms.*

**Response:**

We appreciate this important observation and revised the manuscript accordingly. We now distinguish external validation across RNA-seq and microarray cohorts from platform-agnostic single-sample deployment, and we moderated statements implying immediate clinical utility. We also clarified that the non-inferiority claim is anchored to the elastic-net Cox framework.

To further assess whether the 50-to-24 gene reduction compromised external discrimination, we added a direct paired comparison between CorePAM and the PAM50-full elastic-net model trained on all 50 candidate genes (49 non-zero coefficients at the selected $\lambda$). In these paired bootstrap analyses, discrimination was similar in TCGA-BRCA and favoured CorePAM in METABRIC, GSE20685, and GSE1456 under both cohort-standardised and frozen-reference scoring (Additional file 18).

We agree that external calibration remained imperfect and that frozen-reference performance was not uniform across platforms; we therefore now frame these findings explicitly as a limitation to transportability, requiring platform-specific recalibration before deployment.

## Note on unsolicited corrections

During final quality-control review, we identified and corrected an incorrect platform assignment for GSE20685 (HG-U133A instead of HG-U133 Plus 2.0 / GPL570). After re-running the affected analyses, CorePAM coverage changed from 22/24 to 24/24, while the primary quantitative conclusions remained materially unchanged (largest change: +0.010 in frozen C-index; meta-analytic HR unchanged). All affected outputs included in the resubmission were regenerated. We also added GSE1456 to the frozen-reference sensitivity analysis for completeness.